Intro to Dimensionality Reduction

CSCI-347 Data Mining

Adiesha Liyana Ralalage

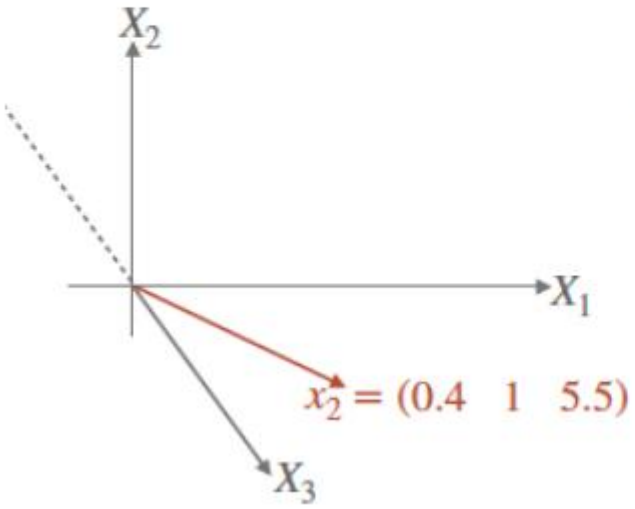
Goal

- We'll explore the concept of high-dimensional space, also known as hyperspace.
- This idea of high dimensionality might be unfamiliar to you.
- Our goal is to become more comfortable with this concept and understand why high-dimensional space may be intuitive as low-dimensional space.

DATA REPRESENTATION IN N-DIMENSIONAL SPACE

Each instance/row/entity can be represented as a vector in m-dimensional space for m-dimensional data

Here, we think of x_2 as a 3-dimensional vector



$$D = \begin{array}{c|ccc} & X_1 & X_2 & X_3 \\ \hline x_1 & 0.2 & 23 & 5.7 \\ x_2 & 0.4 & 1 & 5.4 \\ x_3 & 1.8 & 0 & 5.2 \\ x_4 & 5.6 & 50 & 5.1 \\ x_5 & -0.5 & 34 & 5.3 \\ x_6 & 0.4 & 19 & 5.4 \\ x_7 & 1.1 & 11 & 5.5 \end{array}$$

Let’s think about problems of having high dimensional dataset

- Intuitions from 2D and 3D space does not always work on high dimensional spaces.
- Volume scales in non-intuitive ways.
- Near and far becomes harder to differentiate.
- Computing distances becomes more computationally expensive.
- As result, algorithms on high-dimensional data becomes slower.

Hyperspace, Hypercube, Hyperball, Hypersphere

- **Hyperspace** refers to a space with more than **three** dimensions. In a mathematical context, it's any Euclidean space $\mathbb{R}^n$ where $n > 3$, generalizing familiar 1D lines, 2D planes, and 3D volumes.
- **Sometimes Hyperspace do not behave like familiar 2- or 3-dimension spaces. Hyperspace can be represented as hypercube.**
- **Hypercube:** Assume the data is centered, and let m denote the maximum attribute value, The data hyperspace can be represented as a hypercube, centered at 0, with all sides of length $l = 2m$.

$$\boldsymbol{H_d(l) = \left\{ x = (x_1 \quad x_2 \quad \cdots \quad x_d)^T : \forall i, x_i \in \left[-\frac{l}{2}, \frac{l}{2} \right] \right\}}$$

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- **Hypercube:** Assume the data is centered, and let m denote the maximum attribute value, The data hyperspace can be represented as a hypercube, centered at 0, with all sides of length $l = 2m$.
- **Assume that data is centered, i.e., $\mu = 0$. Let r be the largest magnitude among all the points: $r = max_i\{\|x_i\|\}$**
- **The data hyperspace can be represented as d-dimensional hyperball centered at 0 with radius r, defined as:**

$$\boxed{B_d = \{x : \|x\| < r\} \;\; or \; B_d(r) = \left\{ x = (x_1, x_2, \dots, x_d) : \sum_{j=1}^{d} x_j^2 \le r^2 \right\}}$$

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- The surface of the hyperball is called a hypersphere, as it consists of all points exactly at distance r from the center of the hyperball.

$$S_d(r) = \{x : \; \|x\| = r\}$$

$$or$$

$$S_d(r) = \left\{ x = (x_1, x_2, \dots, x_d) \mid \sum_{j=1}^{d} x_j^2 = r^2 \right\}$$

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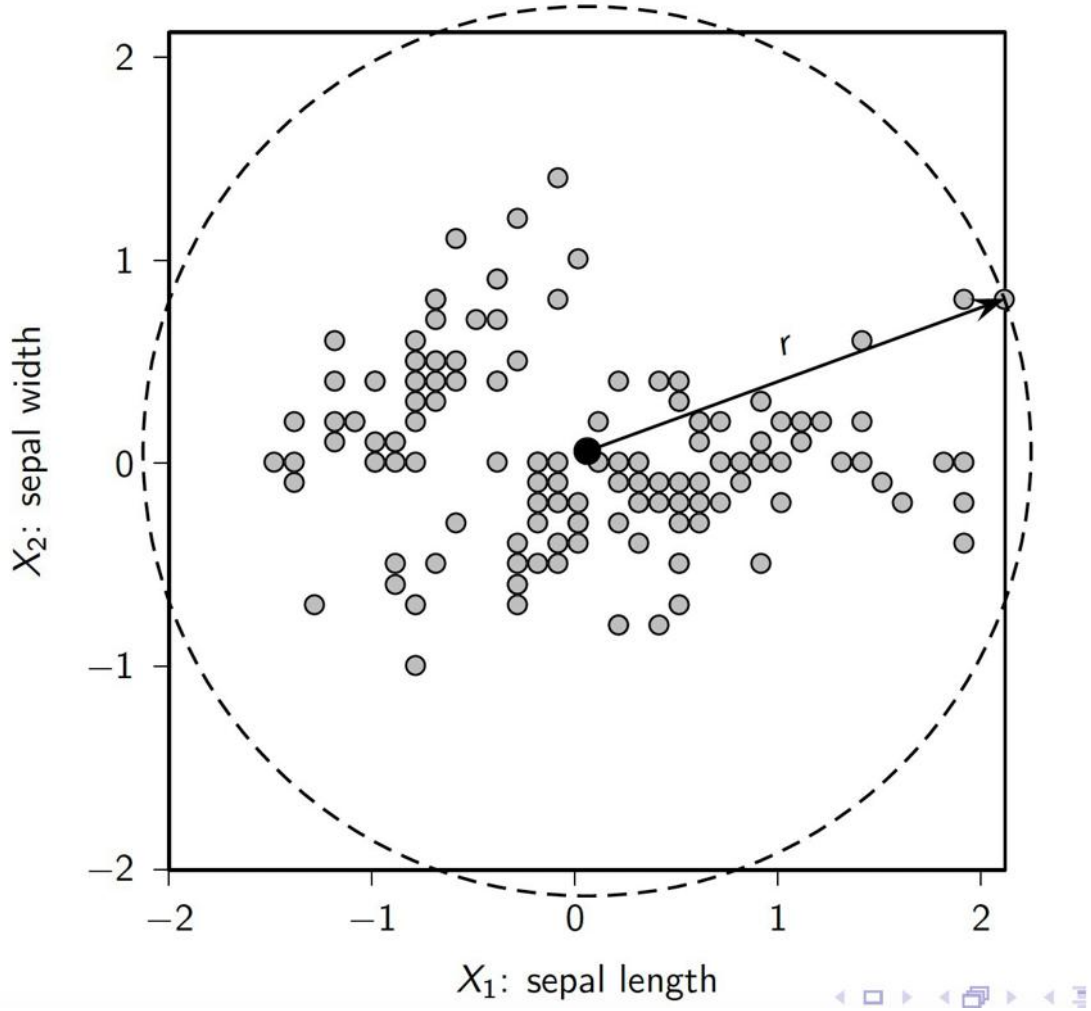
IRIS DATA HYPERSPACE: HYPERCUBE AND HYPERSPHERE



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What happens in high dimensions?

- It gets weird when it goes to higher dimensions.
- Our intuition for 2- and 3-dimensional space doesn't always help us analyze data in high dimensions.
- Example: Volume of a sphere.
  - In 2-dimensions, the volume of a sphere is πr^2.
  - In 3-dimensions, the volume of a sphere is $\frac{4}{3}\pi r^3$
  - In d-dimensions, the volume of a sphere is ……

$$vol(S_d(r)) = K_d \cdot r^d = \left(\frac{\pi^{\frac{d}{2}}}{\Gamma\left(\frac{d}{2}+1\right)}\right) r^d$$

$$\Gamma\left(\frac{d}{2}+1\right) = \begin{cases} \left(\frac{d}{2}\right)! & if\ d\ is\ even \\ \sqrt{\pi}\left(\frac{d!!}{2^{\left(\frac{d+1}{2}\right)}}\right) & if\ d\ is\ odd \end{cases}$$

Euler's gamma function

What happens in high dimensions?

- Let's do a in-class activity: Compute the volume of a sphere with radius 1 in 4 dimensions and 12 dimensions.

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$$\Rightarrow \left(\frac{\pi^{\frac{4}{2}}}{\Gamma\left(\frac{4}{2}+1\right)}\right) 1^4 = \left(\frac{\pi^2}{\Gamma(3)}\right) 1 = \left(\frac{\pi^2}{2!}\right) = \left(\frac{\pi^2}{2}\right) = 4.934802\ (d = 4)$$

$$\Rightarrow \left(\frac{\pi^{\frac{12}{2}}}{\Gamma\left(\frac{12}{2}+1\right)}\right) 1^{12} = \left(\frac{\pi^6}{\Gamma(7)}\right) 1 = \left(\frac{\pi^6}{6!}\right) = \left(\frac{\pi^6}{720}\right)$$

$$= 1.335263\ (d = 12)$$

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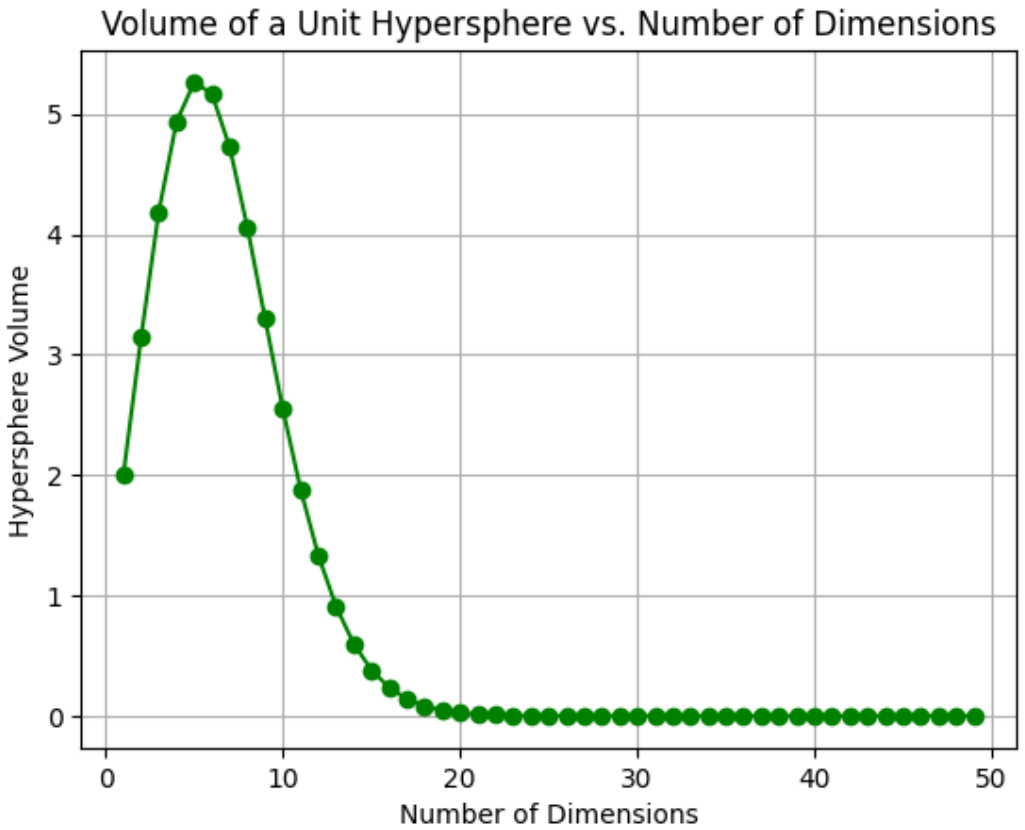
As dimensionality increases, the volume increases up to a point then decreases again.

Let's look at an experiment.

What happens in high dimensions?

- In d-dimensions, the volume of a sphere increases up to a point, then decreases.



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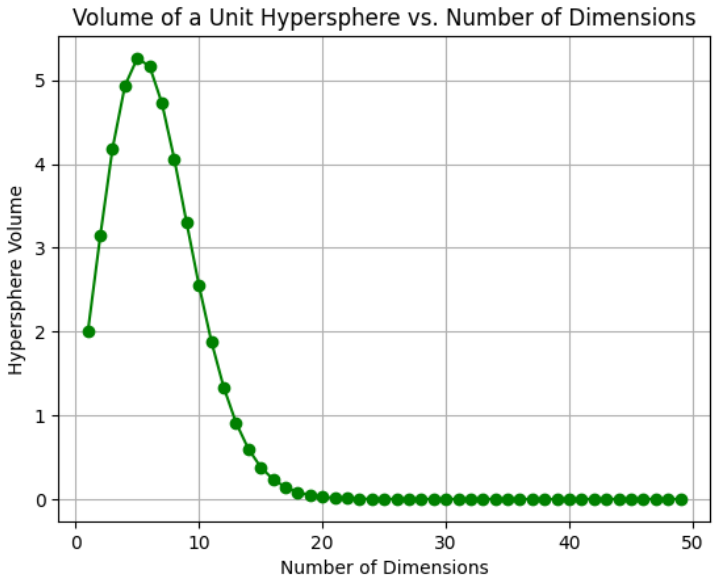
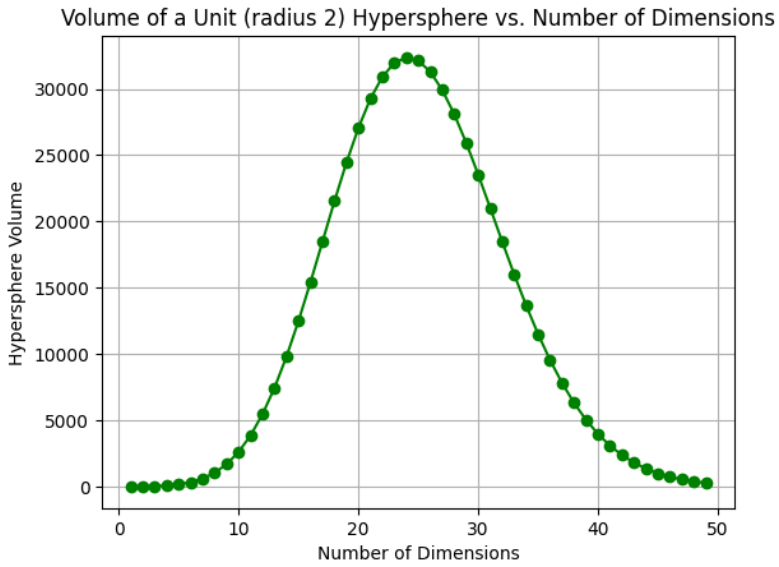
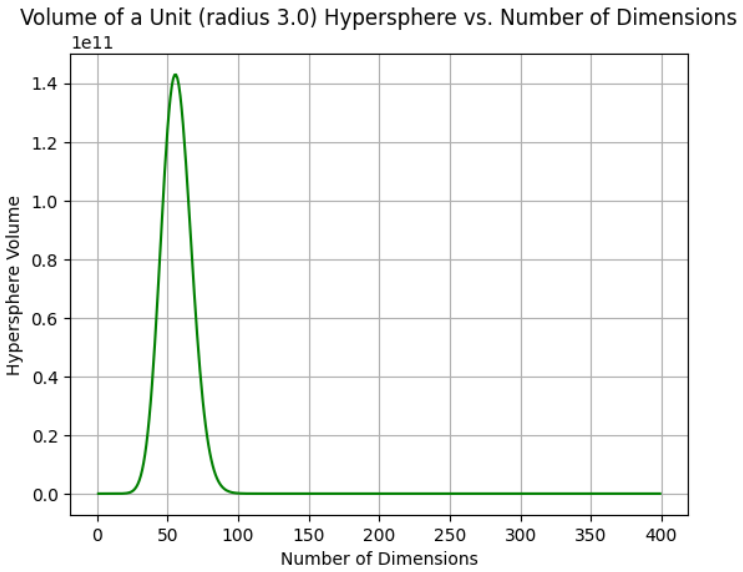






What happens in high dimensions?

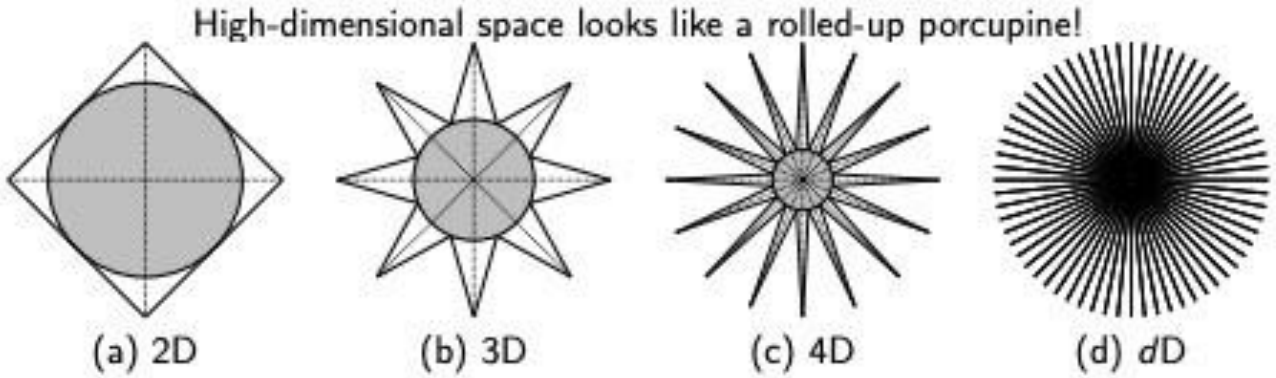
- In d-dimensions, the volume of a sphere increases up to a point, then decreases.
- What if we consider the volume of largest hypersphere inscribed within a hypercube?

$$\frac{vol\left(S_d(r)\right)}{vol\left(H_d(2r)\right)} = \frac{\pi^{\frac{d}{2}}}{\Gamma\left(\frac{d}{2}+1\right)} \cdot \frac{1}{2^d}$$

$$\lim_{d\to\infty} \frac{vol\left(S_d(r)\right)}{vol\left(H_d(2r)\right)} = \lim_{d\to\infty} \frac{\pi^{\frac{d}{2}}}{\Gamma\left(\frac{d}{2}+1\right)} \cdot \frac{1}{2^d} \to 0$$

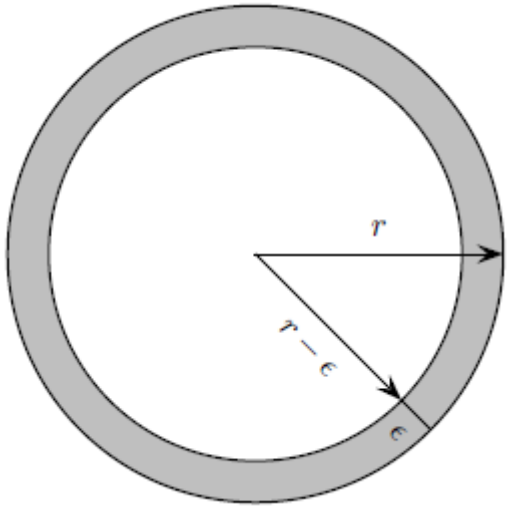
What happens in high dimensions?

- In d-dimensions, the volume of a sphere increases up to a point, then decreases.
- What if we consider the volume of largest hypersphere inscribed within a hypercube?
- As the dimensionality increases, most of the volume of the hypercube is in the "corners," whereas the center is essentially empty.



What happens in high dimensions?

- Let's consider volume of a thin hypersphere shell.
- Consider thin hypersphere shell of width ϵ bounded by an outer hypersphere of radius r.
- Volume of this thin hypersphere is given by the difference between the volumes of the two bounding hyperspheres.

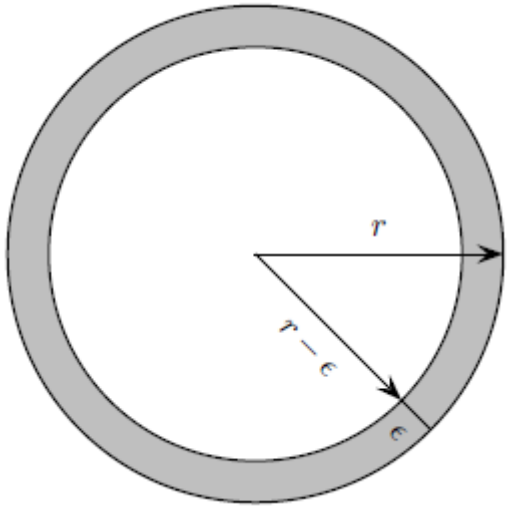


What happens in high dimensions?

- Let $S_d(r, \epsilon)$ denote the thin hypershell of width ϵ.

$$Vol\big(S_d(r,\epsilon)\big) = Vol\big(S_d(r)\big) - Vol\big(S_d(r-\epsilon)\big)$$

$$\frac{Vol\big(S_d(r,\epsilon)\big)}{Vol\big(S_d(r)\big)} = 1 - \left(1 - \frac{\epsilon}{r}\right)^d$$



What happens in high dimensions?

- Let's do an example. Calculate the ratio of volume of the shell vs volume of the hypersphere for radius 1 and $\epsilon = 0.01$ in 2-dimensions and 3 dimensions.

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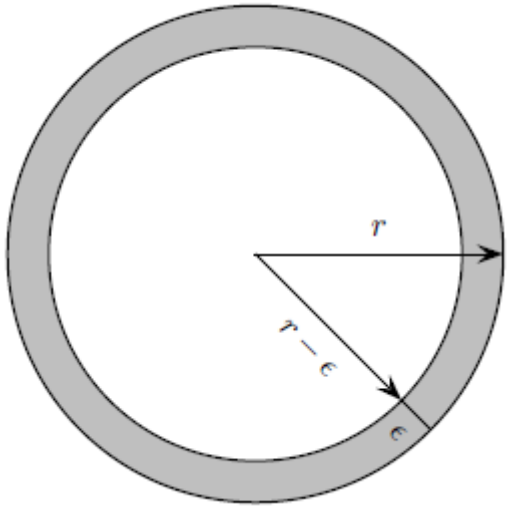

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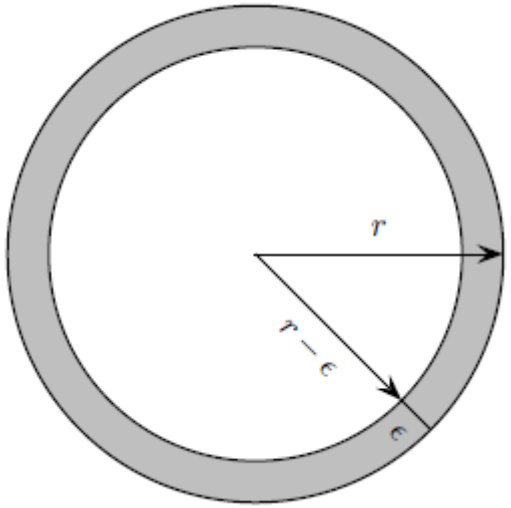
- $r = 1, \epsilon = 0.01, d = 2, \frac{Vol\big(S_d(r,\epsilon)\big)}{Vol\big(S_d(r)\big)} = 0.019 \approx 2\%$
- $r = 1, \epsilon = 0.01, d = 3 \frac{Vol\big(S_d(r,\epsilon)\big)}{Vol\big(S_d(r)\big)} = 0.0297 \approx 3\%$



What happens in high dimensions?

- What happens to this ratio as $d \to \infty$?

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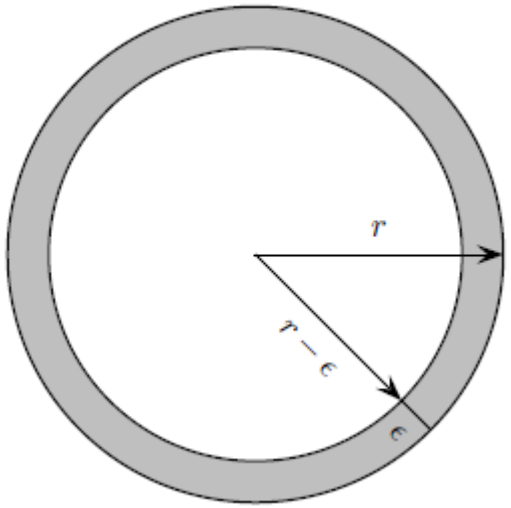
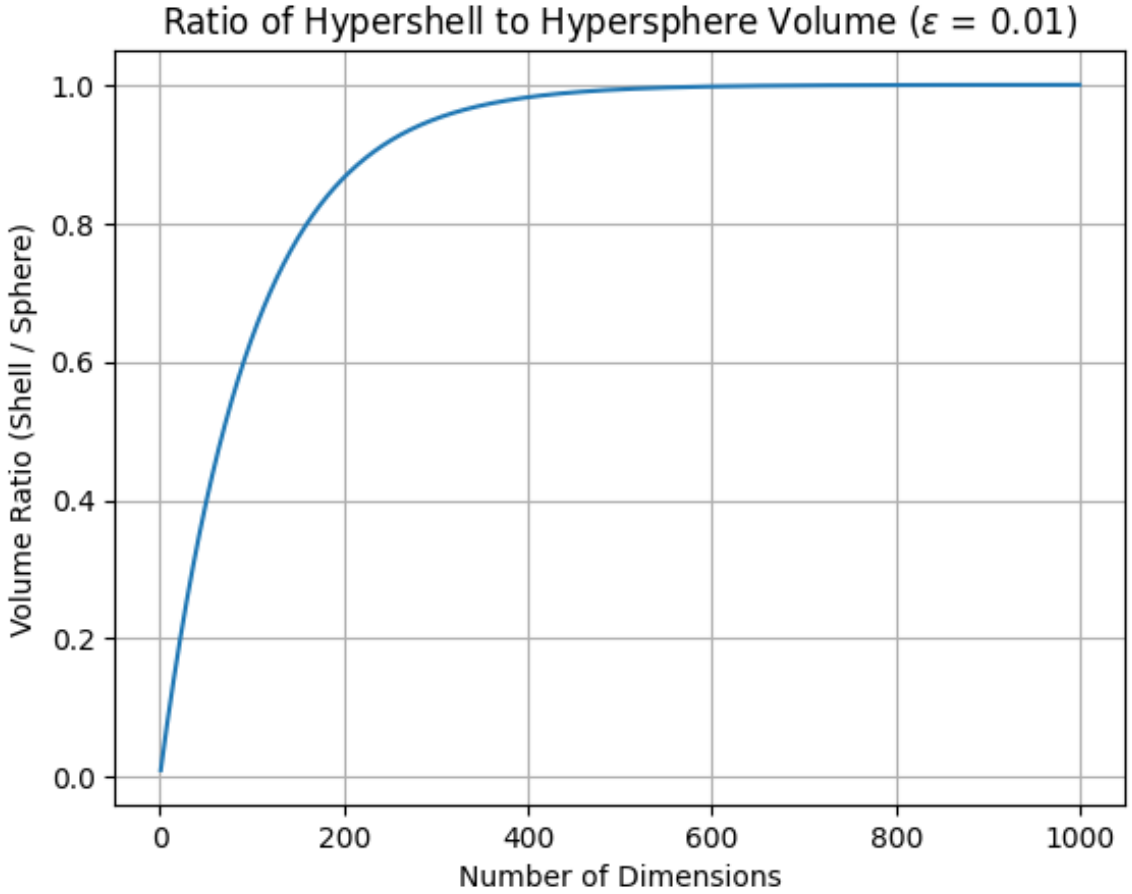


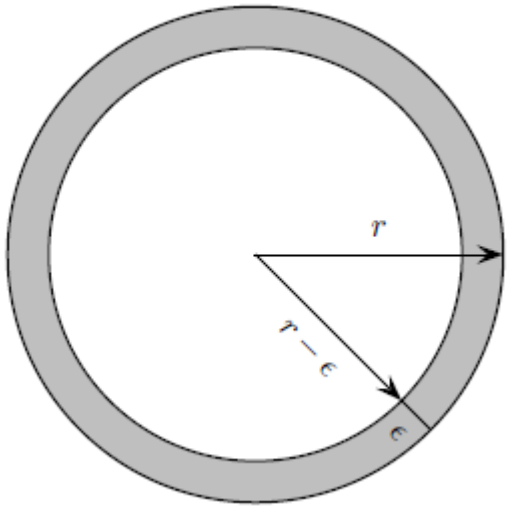
- $\lim\limits_{\boldsymbol{d}\to\infty} \frac{Vol\big(S_d(r,\epsilon)\big)}{Vol\big(S_d(r)\big)} = \lim\limits_{\boldsymbol{d}\to\infty} 1 - \left(1 - \frac{\epsilon}{r}\right)^d \to 1$
- Basically, when $d \to \infty$ almost all of the volume of the hypersphere is contained in the thin shell.
- This means that in high-dimensional spaces, unlike in lower dimensions, most of the volume is concentrated around the surface (within ϵ) of the hypersphere, and the center is essentially void.

What happens in high dimensions?

- What happen

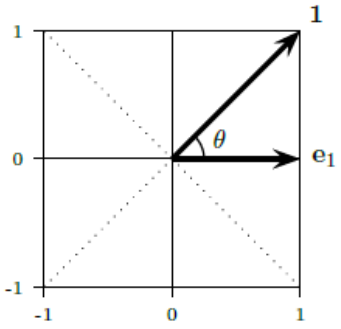
$$\frac{Vo}{V}$$

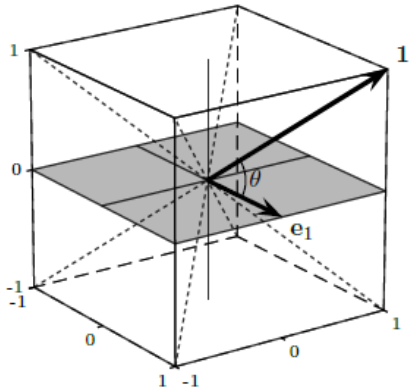




What happens in high dimensions?

- Let's consider diagonals in hyperspace.
  - Another non-intuitive phenomenon.
- Let us assume that we have a d-dimensional hypercube.
  - $origin: 0 = (0_1, 0_2, \dots, 0_d)$, and bounded in each dimension in the range [1, -1].
  - Each corner of this hypercube is d-dimensional vector in the form of $(\pm 1_1, \pm 1_2, \dots, \pm 1_d)$
  - Let $e_i = (0_1, \dots, 1_i, \dots, 0_d)^T$ be the unit vector in dimension i.
  - Let 1 denote the d-dimensional diagonal vector $1 = (1_1, 1_2, \dots, 1_d)^T$





What happens in high dimensions?

- Consider the angle θ_d between diagonal vector 1 and the first axis e_1, in d-dimensions.

$$\cos(\theta_d) = \frac{e_1^T 1}{\|e_1\|\|1\|} = \frac{e_1^T 1}{\sqrt{e_1^T e_1}\sqrt{1^T 1}} = \frac{1}{\sqrt{1}\sqrt{d}} = \frac{1}{\sqrt{d}}$$

$$d = 2, \cos\theta_2 = \frac{1}{\sqrt{2}}$$

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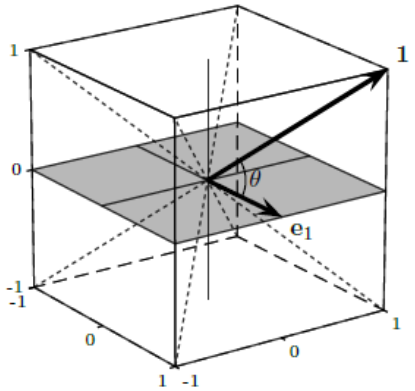

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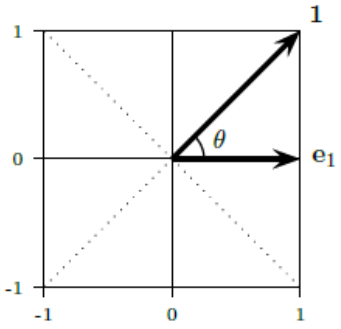
What happens to this when $d \to \infty$?

$$\lim_{d\to\infty} \cos\theta_d = \lim_{d\to\infty} \frac{1}{\sqrt{d}} \to 0$$

$$\lim_{d\to\infty} \theta_d \to \frac{\pi}{2}$$



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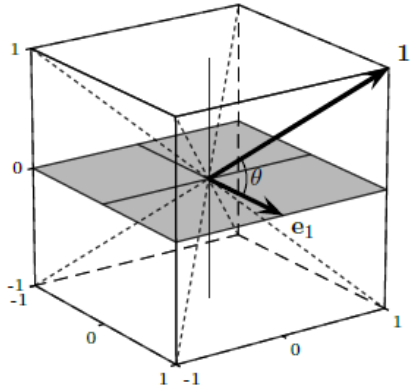
- Consider the angle θ_d between diagonal vector 1 and the first axis e_1, in d-dimensions.

$$\lim_{n\to\infty} \theta_d \to \frac{\pi}{2}$$

- This implies that in high dimensions all of the diagonal vectors are perpendicular (or orthogonal) to all the coordinates axes!
- Since there are 2^d corners in a d-dimensional hyperspace, there are 2^d diagonal vectors from the origin to each of the corners.
- High-dimensional space has an exponential number of orthogonal "axes".
- consequence of this property of high-dimensional space is that if there is a point or a group of points, say a cluster of interest, near a diagonal, these points will get projected into the origin and will not be visible in lower dimensional projections.



What happens in high dimensions?

- As dimensionality increases, the volume of a sphere with radius 1 increases up to a point and vanishes as the dimensionality increases, most of the volume of the hypercube is in the "corners," whereas the center is essentially empty. (Sparsity)
- In high dimensions all the diagonal vectors are perpendicular (or orthogonal) to all the coordinates axes! (Geometric Intuition Breakdown)
- The distance between the nearest neighbor of a point and the farthest neighbor of a point get closer together as dimensionality increases, making it difficult to find true "nearest neighbors" .(Distance Concentration)
  - Check this experimentally.
- This is called Curse of dimensionality

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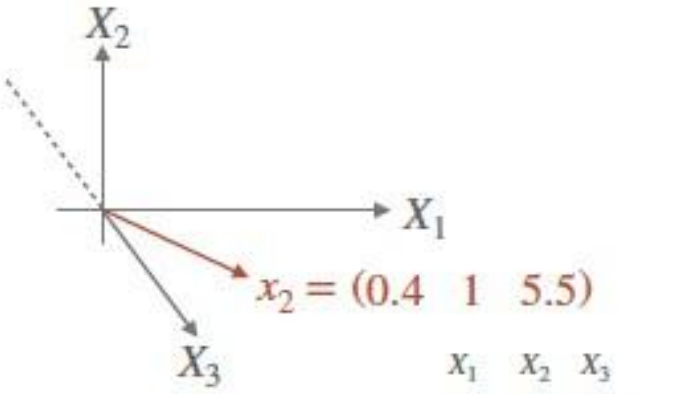
How can we find the best low-dimensional representation of the data?

- How can we reduce the dimensionality without losing information?

How can we find the best low-dimensional representation of the data?

- Dimensionality reduction can be accomplished by:
  - Selecting a subset of attributes of the dataset.
  - What are the possible advantages and disadvantages?



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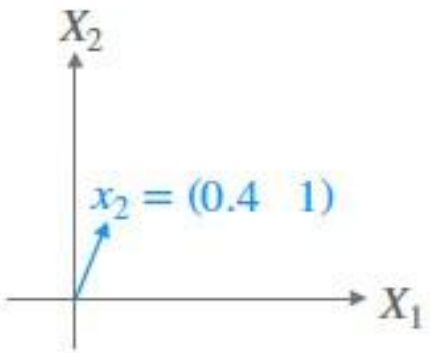


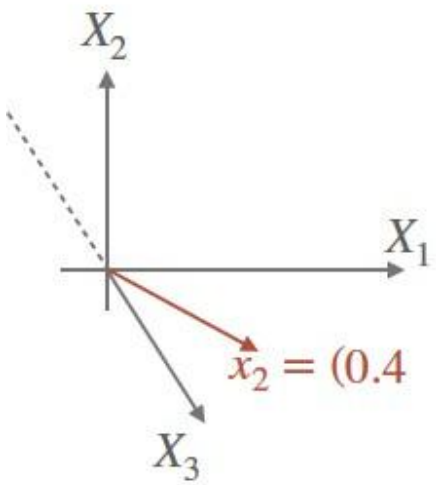
Ideas to reduce dimensionality

Selecting a subset of attributes of the dataset

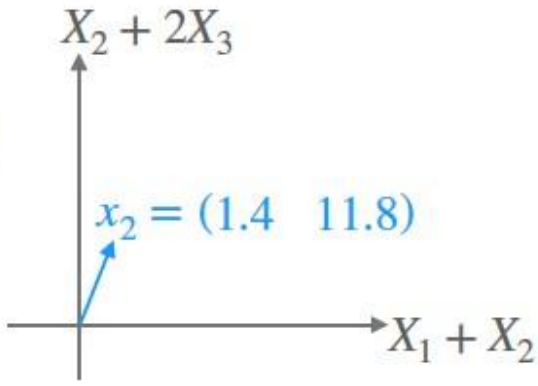
- Advantages
  - Reduced Complexity
  - Improved Model Performance
  - Enhanced Data Visualization
  - Reduced Storage Requirements
  - Increased Interpretability
- Disadvantages
  - Potential Loss of Information
    - Removing features can lead to the loss of valuable information, which may negatively impact model performance.
    - It is crucial to carefully select features to minimize information loss.
  - Risk of Bias
    - If the feature selection process is biased, it can lead to a model that is not representative of the underlying data.

How can we find the best low-dimensional representation of the data?

- Dimensionality reduction can be accomplished by:
  - Selecting a subset of attributes of the dataset.
    - What are the possible advantages and disadvantages?
  - Using all the attributes to create new attributes that can represent the data



$$D = \begin{array}{c|ccc} & X_1 & X_2 & X_3 \\ \hline x_1 & 0.2 & 23 & 5.7 \\ x_2 & 0.4 & 1 & 5.4 \\ x_3 & 1.8 & 0 & 5.2 \\ x_4 & 5.6 & 50 & 5.1 \\ x_5 & -0.5 & 34 & 5.3 \\ x_6 & 0.4 & 19 & 5.4 \\ x_7 & 1.1 & 11 & 5.5 \end{array} \rightarrow \begin{array}{c|cc} & X_1+X_2 & X_2+2X_3 \\ \hline x_1 & 23.2 & 34.4 \\ x_2 & 1.4 & 11.8 \\ x_3 & 1.8 & 10.4 \\ x_4 & 55.6 & 60.2 \\ x_5 & 33.5 & 44.6 \\ x_6 & 19.4 & 29.8 \\ x_7 & 12.1 & 22 \end{array}$$



Ideas to reduce dimensionality

Using all the attributes to create new attributes that can represent the data

- Advantages
  - Capturing Complex Relationships
    - May reveal hidden relationships
  - Increased Model Accuracy
  - Domain Knowledge Integration
- Disadvantages
  - Increased Complexity
  - Increased Computational Cost